

Colony demographics shape nest construction in ants

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eLife Assessment

This study presents an **important** finding that ant nest structure and digging behavior depend on ant age demographics for a ground-dwelling ant species (*Camponotus fellah*). By asking whether ants employ age-polyethism in excavation, the authors address a long-standing question about how individuals in collectives determine the overall state of the task they must perform, and their results may prove to be a key consideration for interpreting results from other studies in the field of social insect behavior. While the experimental evidence that the age of the ants and the group composition affect the digging of tunnels is **solid**, some of the analyses and modeling are seem superfluous, as they do not further support the results or contribute to a deeper understanding of the system.

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Abstract

The ant nest serves as the skeleton of the ant superorganism. Similar to a skeleton, the nest expands as the colony grows and requires repair after catastrophic events. We experimentally compared nest excavation by colonies seeded from a single mated queen and then allowed to grow for six months, to excavation triggered by a catastrophic event in colonies with fixed demographics. The areas excavated by equal group sizes differed significantly between these conditions: heterogeneous populations in naturally growing colonies as well as cohorts of young ants dig larger areas than old ant cohorts. Moreover, we find that younger ants tend to dig slanted tunnels while older ants dig straight down. This is a novel form of age polyethism, where an ant's age dictates not only her likelihood to engage in a task but also the way she performs the task. We further present a quantitative model that predicts that under normal growth, digging is predominantly performed by the younger ants while after a catastrophe all ants dig to restore lost nest volume. The fact that the nests of naturally growing colonies exhibit slanted tunnels strengthens this prediction. Finally, our results indicate how a colony's demographic and physical history are sketched into the current structure of its nest.

Introduction

Nests occur across a range of biological scales and serve various functions such as providing refuge, storage, transport, mating, and micro-climate regulation (1–10). Among the diversity of nests those constructed by social insects are highly complex in terms of function, architecture, and construction mechanisms. These nests are created through digging or modifying a substrate and can be arboreal or underground (11–18). Previous studies have demonstrated that these structures emerge from the interaction between individual workers and the environment in a decentralized manner, showcasing an exemplary instance of biological self-organization (19–28).

Previous studies on nest construction and dynamics often addressed mature colonies of fixed size and over short time scales (24, 29–31). They show that for ant groups of fixed size nest volume grows logistically to a saturation volume that scales linearly with population size (24, 29, 30, 32–34). This led to the hypothesis that workers can perceive the density of ants within the nest and adjust their digging activity accordingly. This mechanism may hold in relatively small populations and simple architectural designs. However, for more complex structures where ants congregate in specific chambers, workers are less likely to assess the overall nest density. In such cases, nest excavation is enhanced by recruiting additional workers from the neighborhood or facilitated by humidity or pheromones (22, 24, 27, 29, 35–38).

Ant colonies are composed of ants that vary in age. It is well known that the task allocation of individual workers can change throughout their lifetimes (age polyethism) (39–46). However, it is unknown which of the work groups in a developing colony participate in the digging effort. Additionally, ant colonies are comprised of the reproductive caste and one or more worker castes that can divide labor among themselves (23, 39, 44, 45, 47, 48). Thus, colony demographics play a significant role in the development of nest structure, and its role needs to be considered.

One way to view the social insect nest is from a colony-superorganism perspective (2, 19, 20, 22, 26). Within this analogy, the social insect nest can be likened to the superorganism's skeleton. Inspired by the processes of skeletal growth and bone healing after fracture we set out to find how the volume of an ant nest is regulated during ordinary colony development and following a catastrophic collapse. Specifically, we were interested in the following questions:

1. How does a colony regulate the co-expansion of the population and the nest as the colony develops from a single queen to tens of workers?
2. How does a colony recover nest volume following a catastrophic collapse?
3. Which ants participate in nest excavation and does this answer differ between the cases (a) and (b)?

We performed two sets of experiments to study how a *Camponotus fellah* ant colony regulates the volume of their nest: The “Colony-Maturation” experiment monitored colonies as they developed from a single fertile queen to a mature colony. We used these experiments to track the coexpansion of the colony and its nest for an extended period (up to over six months). Whereas, in the “Fixed-Demographics” experiments, we monitored excavation by an ant group of determined size and age. We used these experiments to study the nest volume recovery following catastrophic events and the role of different age groups in the digging process.

Results

Nest design and experimental setup

All experiments were conducted within two-dimensional nests (**Fig. 1** [↗](#), Materials and methods), nest sizes are reported as areas but these can be readily translated into volumes by multiplying by the thickness of the setup (0.8 cm). We performed two kinds of excavation experiments: *colony maturation* experiments that follow nest excavation alongside colony growth and *fixed demographics* experiments that track digging by already mature colonies.

We initiated colony maturation experiments by introducing a single mated queen and several brood items (pupae) into the setup. This allowed us to capture nest excavation and colony development from their very first stages. Experiments were terminated either when the queen died or when she became irreversibly trapped after a structural collapse. Overall, colony maturation experiments ($n = 22$) spanned up to 6 months (see Table: Experimental details, Materials, and methods) during which the population size and nest area were recorded. The number of experiments decreased with time: for example, on day 130, we only had 13 ongoing excavation experiments ($n = 9$, collapses).

Fixed demographics experiments ($n = 37$) included colonies with known age demographics. In preparation for these experiments, we tracked the age of individual workers over the course of 10 months (Materials and Methods). We then extracted the queen alongside several workers of a given age and transferred them into the digging setup. Fixed demographics experiments spanned six conditions which varied by worker age (young vs. old workers) and group size (5, 10, & 15 ants, including the queen). These experiments allowed us to track nest excavation as the ants were introduced into the setup and to further quantify the response to an induced catastrophic collapse.

Nest and colony development from the founding queen

Colony maturation experiments allowed us to track the concurrent growth of the colony and its nest.

In all colonies, population size increased logistically (**Fig. 2 A** [↗](#)) and saturated at a value that was highly variable across colonies (typically between 5 and 20 ants, see **SI. Fig 1** [↗](#) and **Table 1** [↗](#)). As the population expanded so did the nest (**Fig. 2 B** [↗](#)). The nest excavation was commenced by the founding queen which dug a mean area of $23.8(\pm 1.7) \text{ cm}^2$ ($n = 22$; unless noted otherwise, means are presented as mean $\pm$ standard error). Once the first workers emerged from the pupae the queen ceased digging (personal observation). However, these first workers did not dig (see kink in **Fig. 2 B-C** [↗](#)) possibly because the initial area dug by the queen led to low ant densities (number of ants/area) at this stage. As more workers emerged the area of the nest increased and varied considerably across nests (**SI. Fig 1** [↗](#)).

For instance, halfway through the experiment duration, around day 95, the excavated area ranged from 38.41 cm^2 to 173.28 cm^2 with a mean excavated area of $100.51(\pm 11.54) \text{ cm}^2$ ($n = 15$). At day 190, excavated areas ranged from 149.87 cm^2 to 234.65 cm^2 with a mean excavated area of $197.47(\pm 13.8) \text{ cm}^2$ ($n = 6$). The number of active experiments decreased over time since experiments were terminated as specified above.

Previous works on already mature colonies revealed a linear relation between population and nest size ([24](#) [↗](#), [29](#) [↗](#)). We find that this relationship persists in cases where colonies and nests are allowed to co-expand (**Fig. 2 C** [↗](#)). A linear fit (R-square - 0.97), indicates that the colonies regulate an area of approximately 11 cm^2 per each ant (**SI Fig. 2 A, B** [↗](#)). Deviations from this fit at early times result from the initial area excavated by the queen.

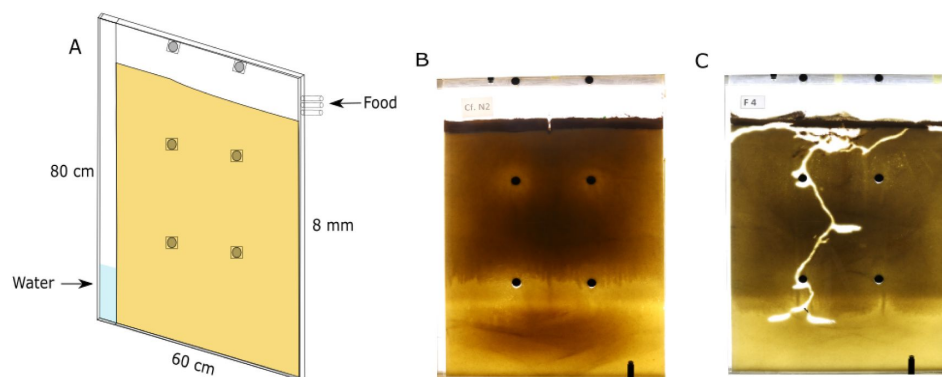


Figure 1.

Experimental system and nest architecture.

(A) We used large nest frames $80 \times 60 \times 0.8 \text{ cm}^3$ to allow for long-term excavation. Nest frames are filled with homogenous fine sand and wet regularly for structural cohesiveness. Water infiltrates through the water column and keeps the sand moist and cohesive. (B and C) A typical nest at the start and termination of an experiment.

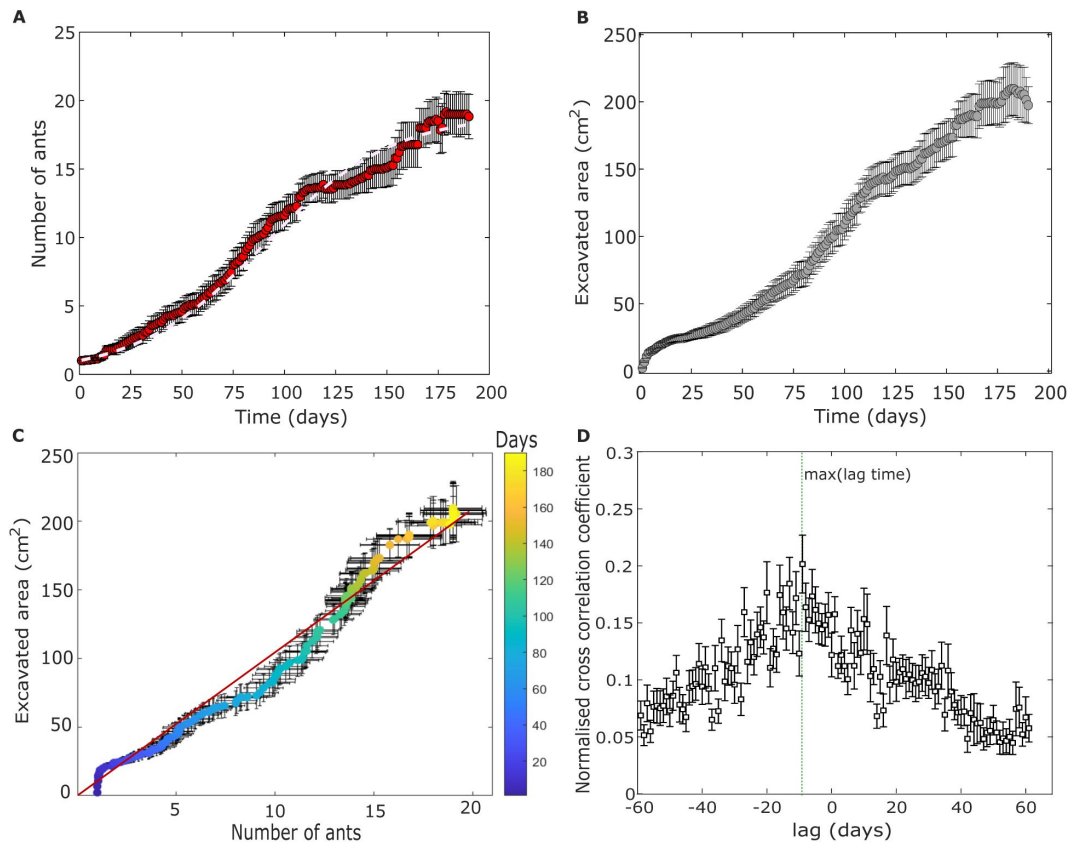


Figure 2.

Excavation and population dynamics during colony maturation.

(A) & (B) Temporal change of the number of ants (A), and excavated area (B) starting from a single queen. Solid red, and grey circles show the mean number of ants and excavated area across the 22 colony maturation experiment. Error bars depicted in black represent the standard error (here, and in all other figures). In (A), we fit a population growth model (white dotted line) using the number of ants as averaged over all colonies (Materials, and methods). (C) The average excavated area is plotted against the corresponding population size. The solid red line depicts a linear fit $y = 10.71x$, $R^2 = 0.97$. The color bar represents the time in days. (D) The normalised cross-correlation coefficient between the digging rate, and population growth rate is shown. The digging rates are correlated with earlier (negative lags) population change with maxima at lag '-10', rather than with the current (zero lag) or future (positive lags). Solid squares correspond to the mean cross-correlation coefficient. The vertical green line highlights the lag time with the maximum mean cross-correlation coefficient.

Colony maturation experiments				
Nest-id	Start date	End date	Max. Popu- lation	Max. Area (cm^2)
F3	23-03-2018	28-10-2018	22	214.44
F4	30-03-2018	28-10-2018	22	291.29
F5	08-04-2018	24-10-2018	16	233.39
F6	05-04-2018	08-10-2018	7	44.35
F7	09-04-2018	08-10-2018	5	21.0
F8	09-04-2018	08-10-2018	8	89.74
F11	12-04-2018	08-10-2018	9	68.68
F12	12-04-2018	24-10-2018	12	154.11
F13	12-04-2018	28-10-2018	20	290.72
F15	18-04-2018	03-10-2018	4	49.67
F16	01-05-2018	24-10-2018	10	183.30
F17	01-05-2018	24-10-2018	9	82.50
F18	01-05-2018	28-10-2018	22	221.39
F19	02-05-2018	24-10-2018	7	161.94
F20	02-05-2018	28-10-2018	18	185.12
F21	10-05-2018	04-09-2018	4	38.72
F22	10-05-2018	24-10-2018	14	121.38
F23	23-05-2018	08-10-2018	4	36.11
F25	31-05-2018	28-10-2018	17	171.02
F26	31-05-2018	28-10-2018	14	145.75
F27	06-05-2018	28-10-2018	11	167.27
F28	06-05-2018	24-10-2018	9	89.78

Table 1.

Details of colony maturation experiments.

All the results stated so far deal with averages over different colonies. These do not allow us to address the causality of the regulation depicted in **Fig 2 C**. To test whether colony expansion triggered changes in nest-area or if it was the other way around, we tracked these concurrent processes one colony at a time. In all experiments, the initial population growth is followed, 15($\pm$ 12) days later, by an increase in excavated area. Moreover, in some of the colonies these dynamics repeat themselves leading to a step-like pattern (**SI Fig. 3**). This staggered pattern suggests that the nest is excavated in separate digging episodes, possibly instigated by a population increase. Cross-correlation analysis between the digging rate and the population growth rate (**Fig. 2 D**) indeed reveals that an increase in the population size is followed by an increase in nest area on a time scale of one week.

Our results suggest that population size controls nest area and not vice-versa. As newly-born ants join the colony the size of the nest increases accordingly. It is, however, not clear which ants do the actual sensing or digging. One straightforward possibility would be that newborn ants are those that dig. Such a mechanism may rid ants of the requirement to determine overall density. Another possibility is that, following the emergence of new ants, every ant, irrespective of her age, senses the increase in density and commences to dig (24, 29, 30).

Fixed demographics experiments reveal age-dependent nest excavation

In “Colony-maturation” experiments ants emerge from their pupae inside the thin confines of the setup. This prevents us from marking them by age. To test how an ant’s age affects her propensity to dig we conducted shorter “Fixed-Demographics” excavation experiments with fixed population sizes (5,10, or 15 ants including a queen) where all workers are of similar age (either “young” (40 $\pm$ 16 days) or “old” 171 $\pm$ 20) (**SI Fig. 4**). These experiments (Materials, and Methods, **Table 2**) further allowed us to study excavation under conditions of extreme area shortage. Under our protocol, such conditions occur twice: as the ants are initially introduced to the setup and then again after we induce a collapse.

Following the introduction of the ants into the setup, the nest area grew logistically and stabilized after about three weeks independently of age or group size (**Fig. 3 A, B**). We found a significant difference in the maximum area excavated by same-sized cohorts as a function of worker age, with the younger ants excavating larger areas when compared to the older ants (**Fig. 3 A, B** - Unpaired Wilcoxon signed-rank test significant for group size: 10; $p = 0.014$, 15; $p = 0.042$). Moreover, the area excavated by the young cohorts was similar to that excavated by naturally maturing colonies at the point in which they reached the same population size (Tukey’s HSD; group size: 5; $p = 0.61$, group size: 10; $p = 0.46$, group size: 15; $p = 0.20$). This holds even after excluding the area excavated by the queen from the colony maturation experiments (**SI. Fig 5**).

Once the fixed-demographics nests reached a steady-state area, we artificially collapsed 25-30 % of the steady-state area (star - **Fig 3A**, & circle - **Fig 3B**). We found that irrespective of age, ants dug after the collapse. Moreover, colonies ceased digging when they recovered (93 $\pm$ 3)% of the area lost by the manual collapse (crossed circles - **Fig 3A**, & **3B**). Saturation areas (**Fig. 3 C**, **SI. Fig 7 A**) increase linearly with the colony size (for the excavated area, General Linear Model (GLM, $F_{(35,37)} >$; $p < 0.0001$). Normalizing saturation areas and digging rates by the number of ants allowed us to obtain the single ant digging rate as well as the target area per ant (this is the reciprocal of ant density which, for short, we refer to as the target area from here on). We find that while the target area of an ant depends on her age (**Fig. 4 A**), digging rates remain stable ($r = 2.2 \pm 0.43 \frac{\text{cm}^2}{\text{ant day}}$, **SI Fig. 6 A, B**), over the first 200 days of an ant’s life (**SI. Fig 7 B**).

Fixed demographics experiments : Young				
Nest-id	Start date	End date	Max. Popu- lation	Max. Area (cm^2)
CF-Y67	28-04-2021	15-06-2021	5	25.15
CF-Y68	28-04-2021	15-06-2021	5	84.27
F-Y35	19-10-2020	24-12-2020	5	42.72
F-Y37	19-10-2020	24-12-2020	5	30.56
CF-Y64	29-04-2021	15-06-2021	10	56.84
CF-Y65	29-04-2021	15-06-2021	10	45.47
F-Y30	19-10-2020	24-12-2020	10	85.74
F-Y31	19-10-2020	24-12-2020	10	92.96
F-Y32	19-10-2020	24-12-2020	10	88.44
F-Y33	19-10-2020	24-12-2020	10	66.93
CF-Y43	10-03-2021	23-05-2021	15	84.98
CF-Y44	08-03-2021	15-05-2021	15	74.92
CF-Y45	08-03-2021	22-05-2021	15	174.83
F-Y43	05-12-2020	31-01-2021	15	132.16
F-Y44	05-12-2020	31-01-2021	15	87.94
F-Y45	05-12-2020	31-01-2021	15	96.03
Fixed demographics experiments : Old				
CF-O69	28-04-2021	15-06-2021	5	27.82
CF-O70	28-04-2021	10-06-2021	5	28.64
CF-O71	30-04-2021	15-06-2021	5	36.21
CF-O72	30-04-2021	15-06-2021	5	15.74
F-O34	19-10-2020	24-12-2020	5	7.65
F-O36	19-10-2020	24-12-2020	5	32.86
CF-O60	28-04-2021	16-06-2021	10	54.65
CF-O61	27-04-2021	15-06-2021	10	18.97
CF-O62	25-04-2021	16-06-2021	10	28.34
CF-O63	28-04-2021	16-06-2021	10	38.13
F-O27	19-10-2020	24-12-2020	10	35.99
F-O28	19-10-2020	24-12-2020	10	35.39
F-O29	19-10-2020	24-12-2020	10	70.62
CF-O46	08-03-2021	16-05-2021	15	76.30
CF-O47	08-03-2021	14-05-2021	15	67.61
CF-O48	08-03-2021	19-05-2021	15	77.39
CF-O50	08-03-2021	20-05-2021	15	95.01
CF-O54	08-03-2021	12-05-2021	15	72.97
F-O40	30-11-2020	31-01-2021	15	37.37
F-O41	30-11-2020	31-01-2021	15	44.07
F-O42	30-11-2020	31-01-2021	15	87.98

Table 2.

Details of fixed demographics experiments.

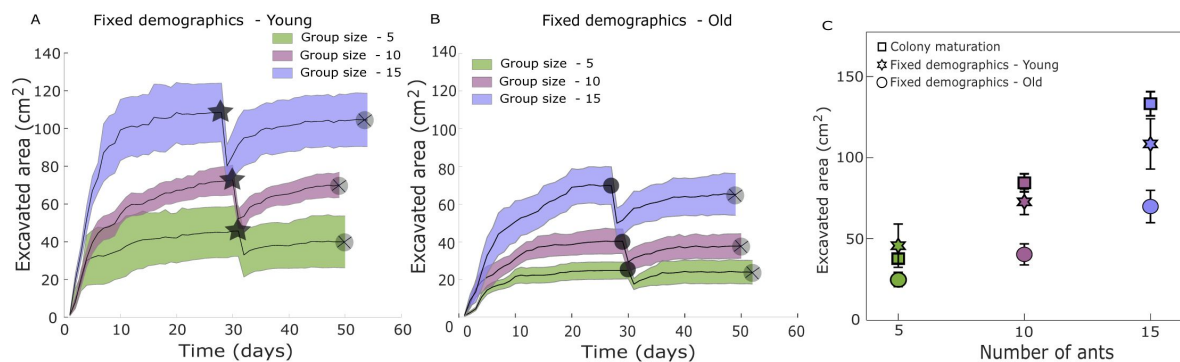


Figure 3.

Excavation dynamics for similar-age groups.

(A & B) The area excavated by the fixed demographics young and old colonies with a colony size of 5, 10, and 15 ants. Solid stars and circles indicate the mean excavated area on the day of collapse. Circular crosses indicate the mean area recovered after the manual collapse. (C) The mean area excavated by colonies of different group sizes from the colony maturation and fixed demographics experiments are shown. Squares (colony maturation), stars (fixed demographics young), and circles (fixed demographics old) correspond to the mean area excavated by colonies of a particular group size. Shaded error bars and error bars correspond to the SEM.

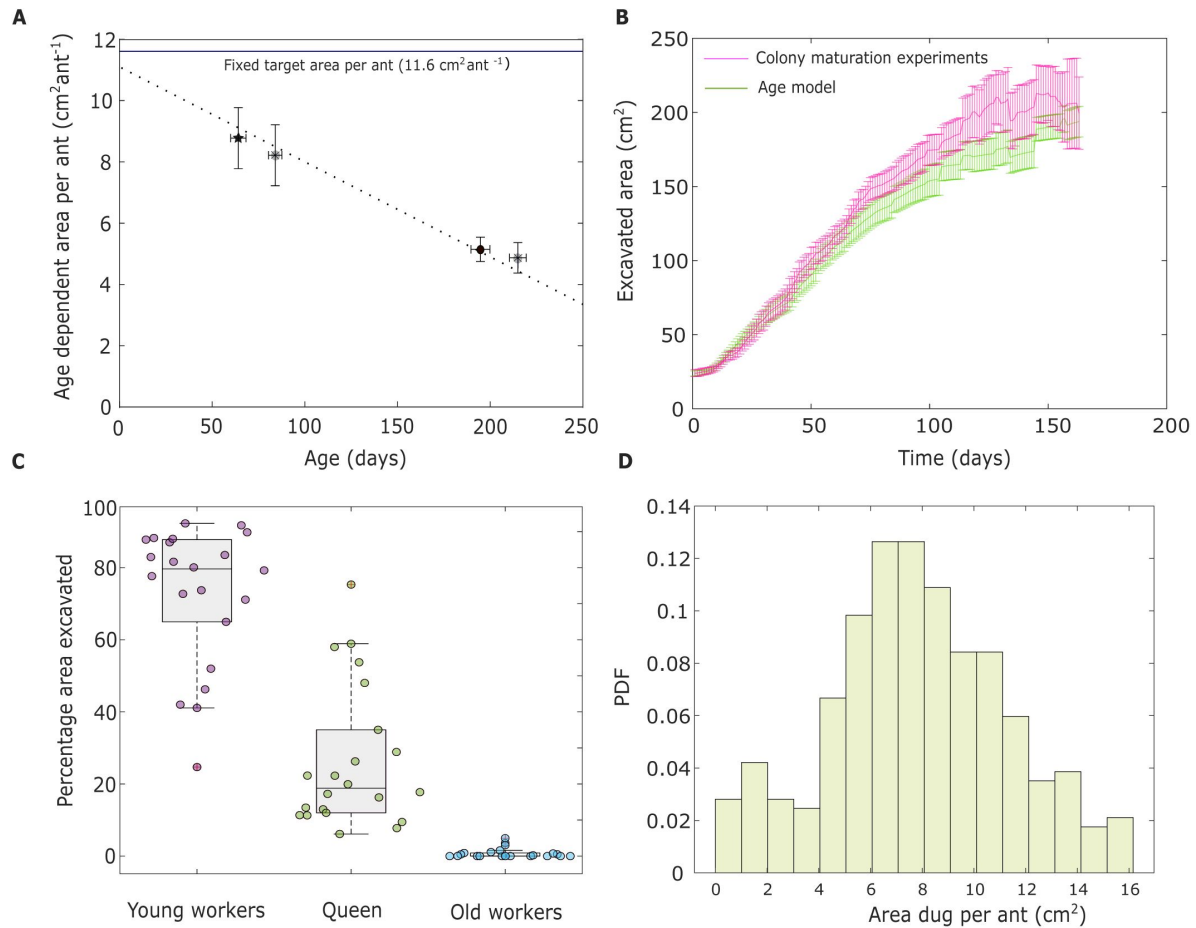


Figure 4.

Age dependent nest excavation.

(A) The target area per ant is represented as a function of the age of individual ants (dotted line). Target areas decrease linearly with ant age ($y = -0.032x + 11.2$, $R^2 = 0.96$). The fixed target area ($11.6 \text{ cm}^2\text{ant}^{-1}$) calculated from the colony maturation experiment is shown with a solid blue line. (B) The mean excavated area as predicted by the age-dependent digging model is shown alongside the excavated area from the colony maturation experiments. Error bars in (A, & B) represent the SEM. (C) Total excavated area during normal colony growth as broken down by young ants, old ants, and the queen as predicted by the age-dependent model. The age threshold for young ants was set to be 56 days. Solid circles represent the area excavated by ants across 22 experiments. Boxes and median lines represent inter-quartile range and median values, and whiskers represent minimum and maximum values of data within 1.5-fold of the inter-quartile range (D) The area excavated by each worker ant during normal colony growth as estimated by the age-dependent model.

Age-dependent model predicts minimal digging by older ants

The fixed-demographics experiments revealed that while all ants dig at the same rate, younger ants aim to approach a higher area target. Building on this, we assume that the age-dependent target area identified in the fixed demographic experiments persists during colony maturation experiments. This assumption facilitates a re-analysis of the colony maturation experiment, allowing us to determine which ants are involved in nest excavation during normal nest growth and investigate its implications.

To do this we present a modified version of a previous model (24, 29, 30) where the digging rate of each ant depends on how similar the current area per ant ($a(t)$ which is one over the density) is to her age-dependent target;

$$\frac{da}{dt} = r \left(1 - \frac{a}{a_{age}} \right) \quad (\text{Eq. 1})$$

Where a_{age} is the target area of a specific age (Fig. 4 A) and the digging rate is $r = 2.2 \pm 0.43 \frac{cm^2}{ant \cdot day}$ (as above). The instances where the current area per ant ('a') is higher than the steady-state area per ant $a_{(age)}$ resulted in a negative digging rate. While there were very few instances where ants filled up the excavated regions they were extremely rare. Therefore, in the model, negative digging rates were taken to be zero.

The total excavation rate dA for a colony that includes 'N' ants of different ages can be obtained by summing over the contributions of all individual ants:

$$\frac{dA}{dt} = \sum_{i=1}^N r \left(1 - \frac{a}{a_{(age)}} \right) \quad (\text{Eq. 2})$$

Since we recorded the population sizes in all colony maturation experiments, and mortality events were rare - we could readily deduce the entire age demographics of the colony for each experiment (SI section: Age quantification from colony maturation experiments, Table 3). Using this data, we could integrate Equation 2, over each experiment to obtain the total excavated area.

The models' results and their comparison to colony maturation experiments are shown in Fig. 4 B. In both model and colony maturation experiments, 'day 0' corresponds to day in which the first workers emerge. The area at this day is the area dug by the queen before this event. We find that the age-dependent model fits the excavation dynamics observed in the colony maturation experiments. Furthermore, we simulated an age-independent density model (SI. Fig 8) with a constant target area for each ant ($11.6 cm^2 ant^{-1}$, SI. Fig 2) as obtained from the colony maturation experiments, rather than an area target that decreases with age (Fig. 4 A). We find that the age-independent model also provides an ample fit for colony maturation experiments, however, it is inconsistent in the light of fixed demographics experiments. An interesting observation is that the target area we fit for the age in the age-independent model closely approximates the empirically measured age-dependent target extrapolated to very young ants (Fig. 4 A). This provides further evidence that, in the colony maturation experiments, it is the youngest ants that do most of the digging.

Since the target area is age-dependent and ants in the colony vary in age, not all ants perform the same amount of digging. By integrating equation 2, we can track the specific terms in the sum and quantify the area excavated by each ant. When setting an age threshold at 56 days (mean + standard deviation of the age of the younger cohort) in the age-dependent model, we find that, after the initial excavation by the queen, younger ants performed the vast majority of the digging (Fig. 4 C). The area excavated by older ants (age of 57 days and above) was nearly negligible. Thus, the age-dependent model provides indirect evidence indicating that younger members of the

Day	Ant Population	New Ants	Demographics
...	...	...	...
10	1		10 days
11	2	1	1 day, 11 days
12	3	1	1 day, 2 days, 12 days
13	3	0	2 days, 3 days, 13 days
...	...	...	...

Table 3.

Colony demographics from nest population.

colony predominantly carry out nest excavation. Furthermore, we quantified the area dug by each ant in the normal colony growth experiment as estimated from the age-dependent model and found that all ants excavated more or less the same amount, with a mean area of $7.84(\pm 3.44) \text{ cm}^2$ (where the error bar represents the standard deviation) (Fig. 4 D; Kolmogorov-Smirnov test, $p = 0.56$).

These findings suggest that while all ants can dig and will do so following catastrophic events (see Fig. 3), the regular nest expansion is primarily carried out by young ants.

Structural classification of nests

The colony maturation and fixed demographics experimental conditions differed not only in the total volume of the nest but also in its architecture. We compared nest architectures by segmenting the nest into chambers and tunnels, where chambers are flat horizontal structures, whereas tunnels are more narrow and vertical (SI Fig. 9, SI section: Nest skeletonization, segmentation and orientation, (13, 30, 49, 50)).

We found that the colony maturation colonies exhibited a “top-heavy” design (13, 51, 52): large chambers (marked in Yellow in Fig. 5 A) comprise most of the nest area at the top part of the nest (52). Tunnels generally branch out from the large chambers and mostly lead to the bottom parts of the nest. The wide tunnels are primarily seen originating from the tunnels and are oriented parallel to the chambers (Fig. 5 A). In the fixed demographics experiments, the nest structure mostly lacked large chambers, unlike the colony maturation experiments (Fig. 5 B, C, D, E, and F). This could be attributed to the absence of a young queen prone to dig and/or to the absence of brood. We analyzed the orientation of all tunnels and chambers relative to the direction of gravity. We found that in the colony maturation and fixed-demographics young experiments, the bulk of tunnels and wide tunnels are oriented between ($30^\circ - 60^\circ$). Surprisingly, in the fixed-demographics old colonies, the orientation of tunnels ranged from ($70^\circ - 90^\circ$) (Fig. 5 G, H; SI Fig. 10). Thus, the tunnels and wide tunnels in the colony maturation experiments are similar to the fixed demographics of young colonies but not the old ones.

Maturing colonies are composed of ants of different ages. Nevertheless, colony maturation experiments displayed tunnel orientation similar to those in the nest of young fixed demographics cohorts. This supports our prediction of the age-dependent model that younger ants primarily undertake digging tasks within maturing colonies.

Discussion

Group size is a key determinant in the behavior of ant colonies. Groups of different sizes differ in their foraging strategies (53–57), division of labor (2, 42, 45–48, 58, 59), and communication networks (60–63). It is less clear what happens in the expansion period that follows colony foundation, when group size is constantly on the rise. A growing colony may exhibit collective patterns that differ from a fixed group of the same size. This is especially relevant if the system preserves long-term memories such as the stigmergic alteration of the environment evident in nest construction (22, 30, 64–67).

To date, the overwhelming majority of studies on nest excavation in ants were short-term experiments performed on a mature group of fixed size (24–26, 29–31, 35, 52, 67). As we have shown, these short term studies likely address construction in emergency scenarios where there is a huge deficit in available space. Studying animal behavior in ethologically relevant conditions is most valuable for being able to correctly capture the nature of

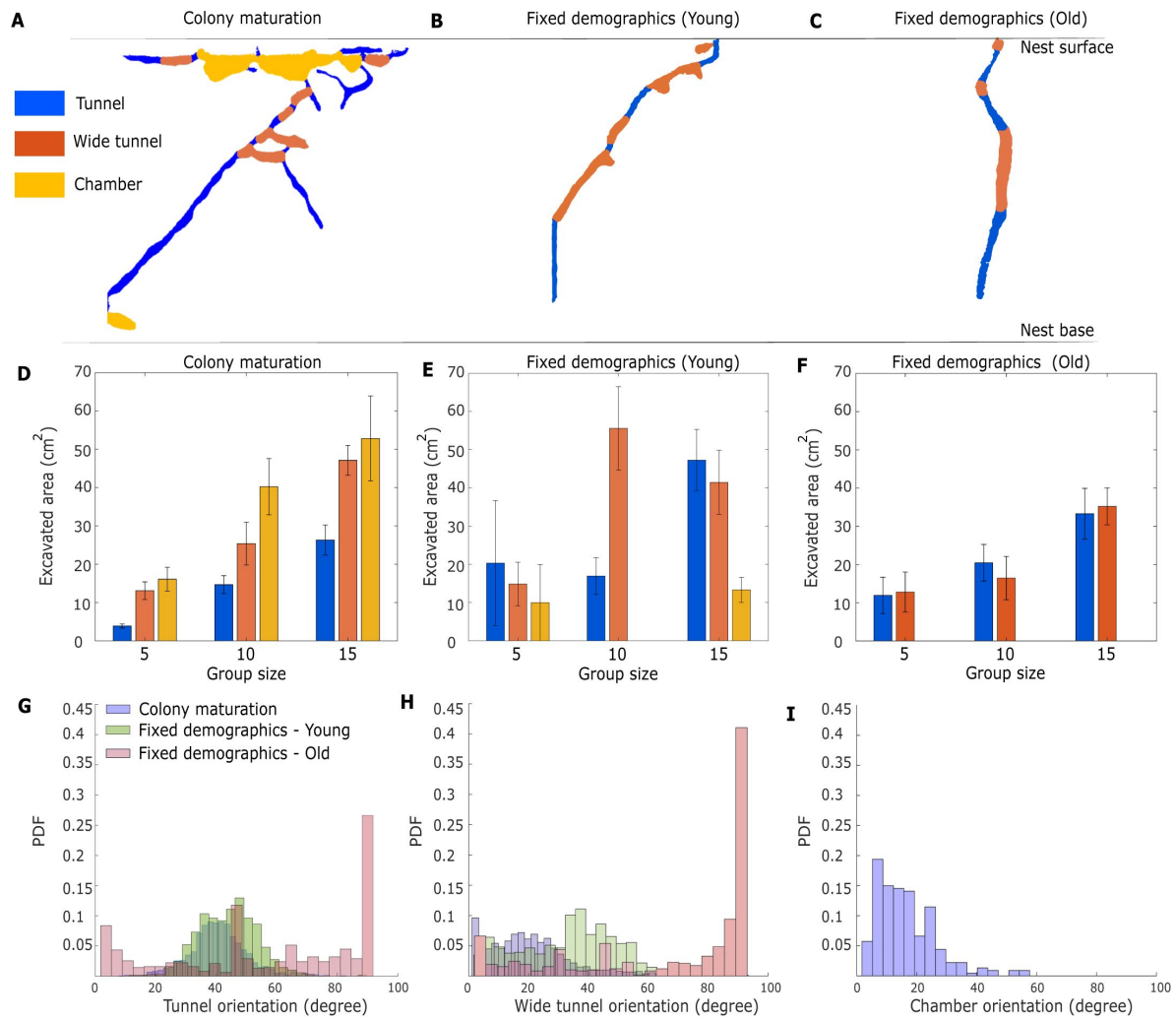


Figure 5.

Structural classification of nests.

(A, B, & C). Panels represent the nests excavated from the colony maturation (A), fixed demographics young (B), and old (C) nests in terms of the tunnel, wide tunnel, and chambers for a colony size 15. (D, E, & F) The contribution of tunnels and chambers to the total excavated area for a fixed group size across all the experimental conditions. Error bars represent the SEM. (G, H,&I) Angular orientation of tunnels and chambers across conditions.

these animals and is crucial for deciphering the underlying meaning of their behavior. Furthermore, testing animals in the wrong context/conditions could lead to misleading conclusions (68–73).

The current study follows ant colonies over extended periods as they mature from a single reproductive individual. We find significant quantitative and qualitative differences between nests constructed within this natural context and nests constructed in the context of an emergency. These include the total volume of the nest, the way this volume is divided into tunnels and chambers, the orientation of the tunnels, and the identity of the ants that do the actual digging.

A main advantage of social insects is their ability to allocate different tasks to different individuals. A well-known example includes age-polyethism wherein ants perform age-specific tasks (2, 42, 45–48, 58, 59). Response thresholds that vary with age have been suggested as a mechanism for age polyethism (23, 74–78). Previous studies of nest excavation indeed rely on a threshold mechanism that prompts ants to dig extra space when the nest becomes too crowded (24, 29, 30, 35). Our results reveal that this area target decreases linearly with age such that young ants are more sensitive to shortages in space.

Age polyethism plays an important role in controlling the volume of the nest. During normal growth conditions, nest volume is maintained proportional to the number of ants, and densities never become too high. This means that density thresholds for older ants are never crossed and excavation is performed almost exclusively by younger ants. This fact has implications for the volume of the nest. A natural growing nest includes ants that have eclosed inside the nest and thus have the highest possible volume per ant target. This is in line with the fact that the volume of such a naturally formed nest is higher than the volume of a nest excavated by an ant group of equal size. Under these conditions, effectively, each new ant joins the colony and adds a small volume to the nest. Following a catastrophic collapse or displacement volume shortages become critical and ants of all ages join the excavation effort.

Variable digging behaviors impact not just the nest's volume but also its structure. During the initial phases of colony establishment, only the queen participates in excavation activities. Our findings reveal that, during this period, the queen typically creates several large chambers near the surface. Aligned with the density-target threshold model, these chambers prolong the emergence of the first worker ants, who eventually take over excavation tasks. Consequently, the queen's initial digging alters the nest's architecture to become more top-heavy. Moreover, we found that an ant's age influences not only its likelihood to engage in a task but also how it performs that task.

We believe that the nest reflects the cumulative effects of past events and conditions experienced by the colony. For example, in line with our experimental findings, if the colony expanded during a period with many young ants, more slanted tunnels are expected, indicating accommodation for a growing population. Conversely, nest construction initiated by mature ants or a catastrophe leading to nest area reduction would result in newly constructed tunnels aligning more with gravity, running straight down. Thus, the nest structure could be treated as a historical record of the colony.

We wish to stress that tunnel orientation and its dependence on the age of the diggers constitute a new kind of age-polyethism. Previously reported examples of age-polyethism regarded it as a means of allocating a given task to different subgroups in the colony (79–85). Our results demonstrate a new kind of behavioral plasticity wherein the task itself, and not just the propensity to do it, drastically changes with the age of the ant. This new kind of age polyethism adds previously unknown regulatory capabilities to the ant colony and is distinct from task performance based on morphological differences seen in polymorphic ants (27, 31).

To conclude, ant colony demographics play a pivotal role in nest excavation. They affect the identity of the ants that do the digging, the dynamics of nest growth, the volume of the nest per population size, and the chamber and tunnel architecture of the nest. We find that age-polyethism provides a major means by which the colony exerts control over its nest in different scenarios.

Materials and methods Study species

We chose to work with *Camponotus fellah*, a common species in Israel of the ground-dwelling carpenter ants (7.8–17.2 mm body length, Ionescu-Hirsch, 2009). These ants are omnivorous, monogynous, and singly mated (86%), with colony sizes ranging from tens to thousands of workers. We collected newly-mated queens in the Weizmann Institute of Science campus after nuptial flights in March 2018. For the colony maturation experiments, we used only queens with pupae, that had already nurtured larvae to increase survival rates. We used colonies of group sizes (5, 10, and 15 ants) for the fixed demographics experiments. We housed the single queen and fixed colonies in plastered nest containers and environmental conditions (climate-controlled chamber under controlled humidity (65%), temperature (27° C), and a light-dark cycle of 12 hours) before the experiments. We supplied the colonies with a food mixture (tuna, honey, eggs, vitamin mix, and water), water, and a 20 % sucrose solution.

Nest structure

The artificial nest structure consists of a 2D Plexiglas frame 60 cm wide, 80 cm high, and 8 mm deep made from 5 mm thick cast acrylic sheets. Six support blocks (2 cm) held the front and back frames together. We equipped the nests with a 5 cm wide water column connected by a series of 0.4 mm holes in the bottom that allow water transfer but not sand grains (Fig. 1 A). The top part of the frame was coated with Fluon (Sorpil) to prevent the escape of ants. A removable plexiglass lid perforated with numerous small (1 mm) holes for ventilation covers the top of the nest.

Nest preparation

Each nest was filled 70 cm high with homogenous fine-grained sand (AccuSand 30 – 40, Unimin Co, USA) with grain sizes of 425–600 µm. Before the experiments began, water infiltrated gradually from the water column to the sand, slowly saturating the entire sand column from the bottom. Apart from wetting the sand, this action rearranged the sand grains in even more compact packaging and lowered the chance of spaces opening up in the column. On top of the sand layer, we placed a 1–2 cm layer of soil on the sand column, thus significantly reducing evaporation and the subsequent collapses. The soil was collected from the Weizmann campus orchard and sieved through coarse (2 mm) and fine (0.5 mm) sieves (Haver & Boecker, Germany), and was sterilized. We manually made a 30 mm long and 8 mm wide hole in the middle of the solid layer (seeding point) to induce a digging behavior.

Experimental overview

The study included 22 colony maturation and 37 fixed demographics experiments (Fixed demographics young, group size 5, n = 4, group size 10, n = 6, group size 15, n = 6); Fixed demographics old (group size 5, n = 6, group size 10, n = 7, group size 15, n = 8) of ant nest excavation. We placed the founding queen on top of a 60 cm wide, 70 cm high, and 8 mm thick soil-sand column in each nest. We photographed the nest development every 1–3 days. terminated the investigations once the queen died or became trapped after a collapse.

Fixed demographics colonies

Queens used for the fixed demographics experiments were collected during the nuptial flight of May 2020. Forty-five newly mated queens were reared for the fixed demographics experiments and 37 colonies that attained the required colony sizes needed for the fixed demographics were chosen for the experiments. The colonies were maintained with a 12-hour light/dark cycle, room

temperature of $25 \pm 2^\circ\text{C}$, and 60% humidity. Each freshly born ant was marked using a unique three-color combination of acrylic paint on the head, thorax, and abdomen to determine the age. We checked each colony at least once in 5 days to check whether new workers emerged. The marked worker's age was recorded and updated in due course. In the fixed demographics experiments, we separated colonies into two different demographics: Young and Old. The young and old colonies had workers with an average age of 40 ± 16 days and 171.56 ± 20 days, respectively. Error represents the standard deviation of the mean.

Experimental conditions

We maintained a stable environmental condition in the experimental room by maintaining a 12-hour light/dark cycle and room temperature of $25 \pm 2^\circ\text{C}$. Daily variations were 2-3 degrees, usually $24 - 27^\circ\text{C}$. More significant deviations ranging at most $20 - 30^\circ\text{C}$ did occur sporadically due to technical problems, but they did not last more than a few hours at a time. Humidity was more challenging to maintain, and it fluctuated in the range of 40 – 70%, averaging around 60%.

Nest maintenance

We administered food for ants ad libitum via three tubes of water, sucrose water (20%), and protein food. The protein consisted of egg powder, tuna, prawns, honey, agar, and vitamins. We filled the water and protein food in 5 ml polystyrene tubes and placed a cotton stopper at the end to prevent overflow. We connected the tubes at a slight angle to ensure the flow of liquids using a custom-made plexiglass adapter. We replenished three tubes (water, sucrose water, and protein food) regularly and performed nest maintenance once every two weeks.

Nest photography - Data collection

A photography sequence of different exposures recorded the nest development in a measurement. The nests were placed one meter away from the camera against a diffuser-covered LED backlight. Images were initially taken with Canon EOS 800D DSLR cameras with a 24 MP resolution and fitted with EF-S 18-55mm f/4-5.6 IS STM lenses. We controlled the cameras remotely using a USB cable through the 'DigiCamControl' software (Istvan, 2019) with the photography sequence programmed using MATLAB CameraController (Serge, 2019). The photography sequence consisted of four photos taken at 2-second intervals.

Image processing and Excavated area

The images were post-processed using the MATLAB Image Processing Toolbox. The first step was image registration, where we aligned nest images with the first image taken before introducing ants. The algorithm registers the images through a projective geometric transformation using the six support blocks in the frame as control points. After registration and cropping, we binarized the images to show only the excavated areas of the sand column. The first binarization step involved subtracting the initial image from the processed image to reduce glare and noise. We then converted the RGB image into an intensity image by choosing the maximum value between the green or blue channels. Next, we used adaptive thresholding to produce BW (black and white) images of the nest architecture in which white pixels represented the excavated areas. Lastly, we made manual corrections to mark known excavated zones, which were not evident in the images, or remove other apparent noises. The excavated area refers to each Black and White image's white pixel area.

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Author contributions

Conceptualization and methodology, all authors; software, validation, formal analysis, investigation, and visualization, all authors; resources, all authors; writing - original draft and review & editing, all authors; supervision and funding acquisition O.F.

Declaration of interests

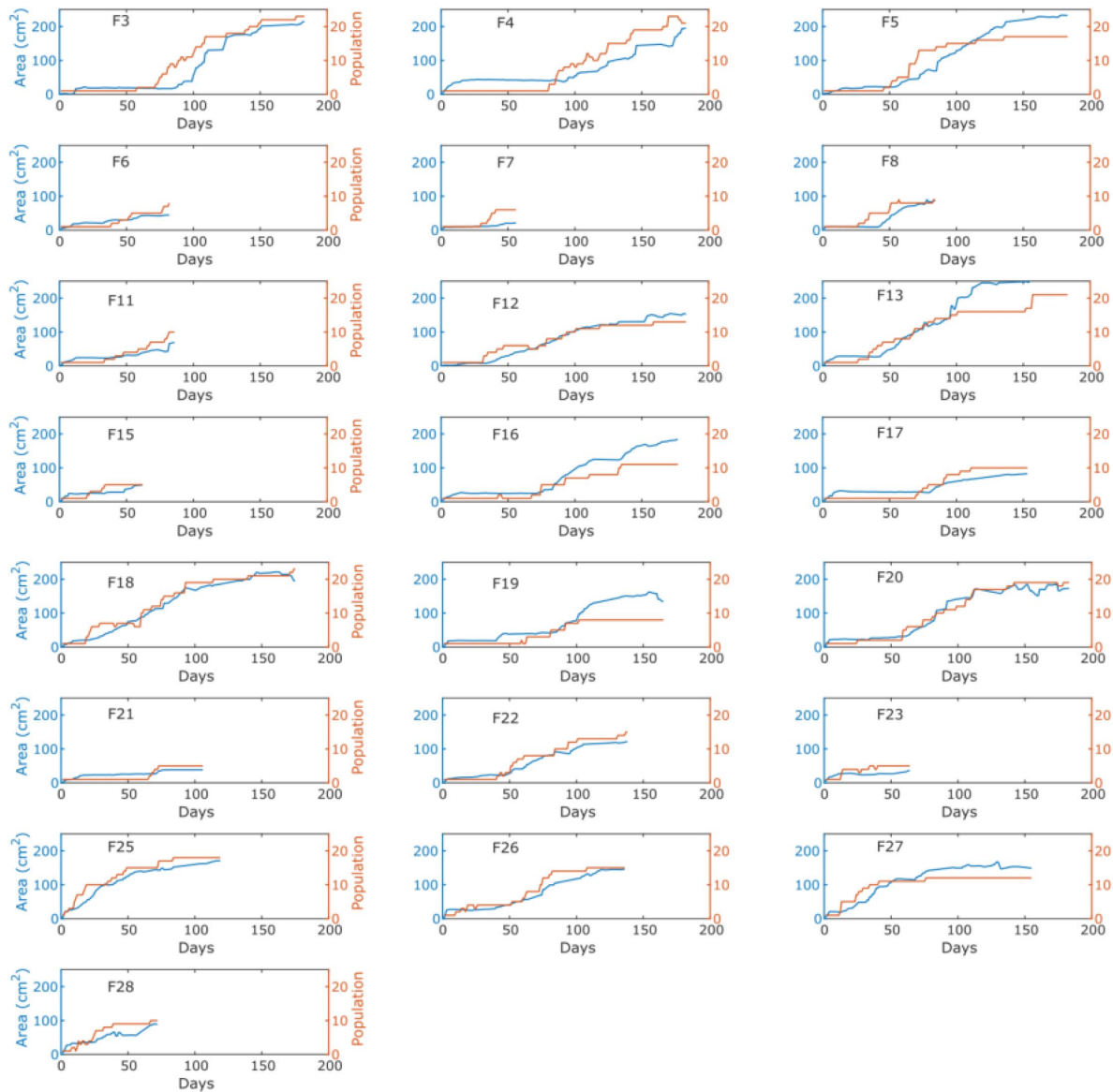
The authors declare no competing interests.

Data availability

All data presented in this manuscript will be made available in a data repository upon acceptance.

Supplementary information

Individual experiments - Colony maturation

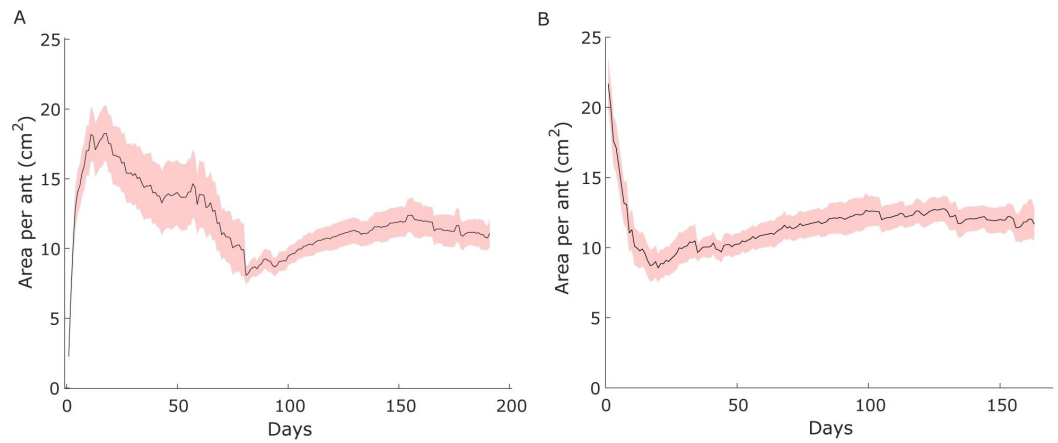


SI Figure 1.

Individual experiments - Area, Population.

The change of area and population across individual colony maturation experiments ($n = 22$) are provided. The left y-axis shows the excavated area and the right y-axis shows the population.

Area per ant from colony maturation experiments

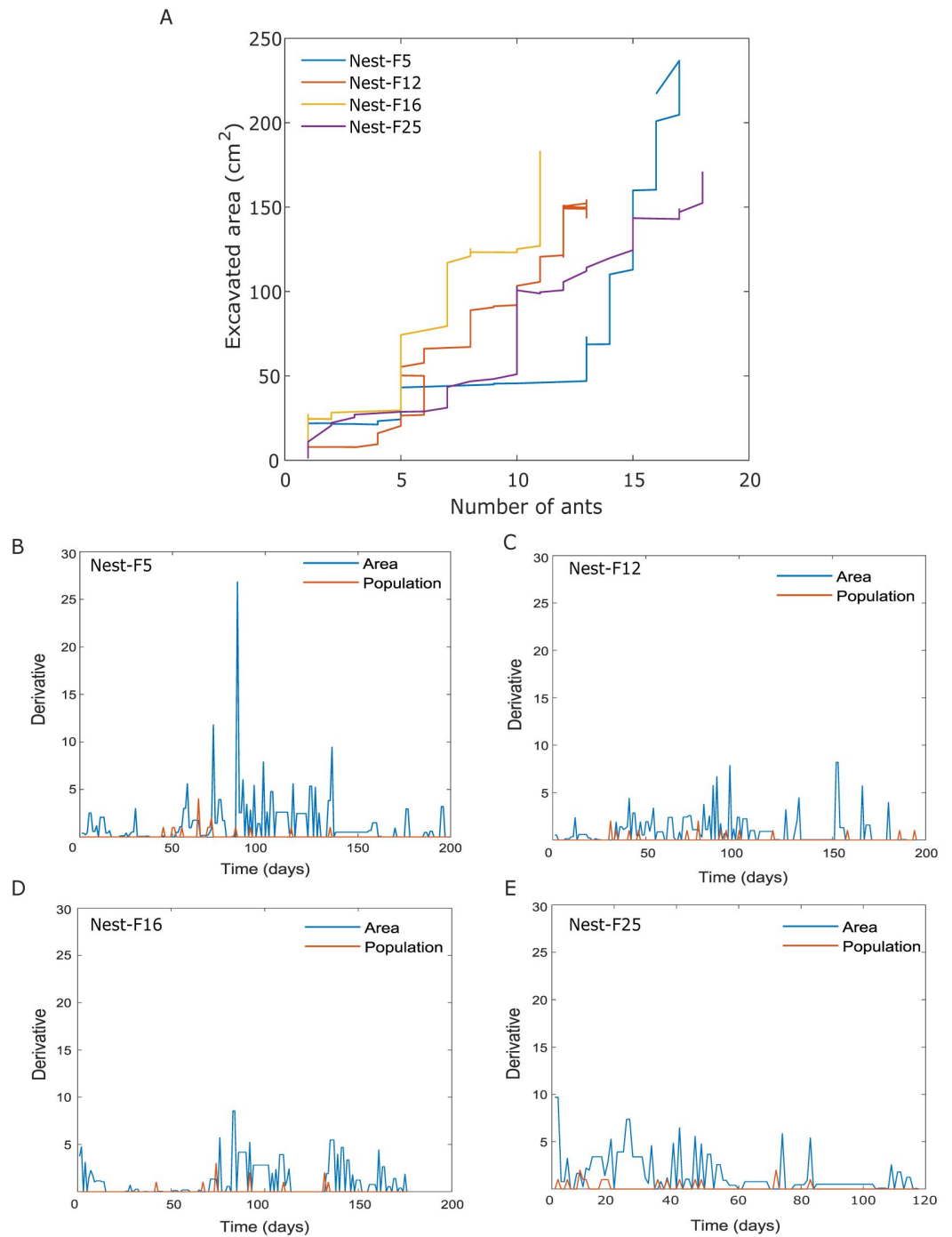


SI Figure 2.

Temporal change of the area per ant.

The temporal change of the area per ant from the colony maturation experiments. The area per ant is calculated by dividing the excavated area by the number of ants in the experiment. In **Fig. 2 A**, day '0' corresponds to the start of the experiment, whereas in **2 B**, day '0' corresponds to the day on which the first workers were born. The area per ant stabilizes at $11.1(\pm 1)$ for **Fig. 2 A**, and $11.6(\pm 1.15)$ for **Fig. 2 B**. Shaded error bar in both figures shows the SEM.

Individual colony maturation experiments reveal a step-like pattern



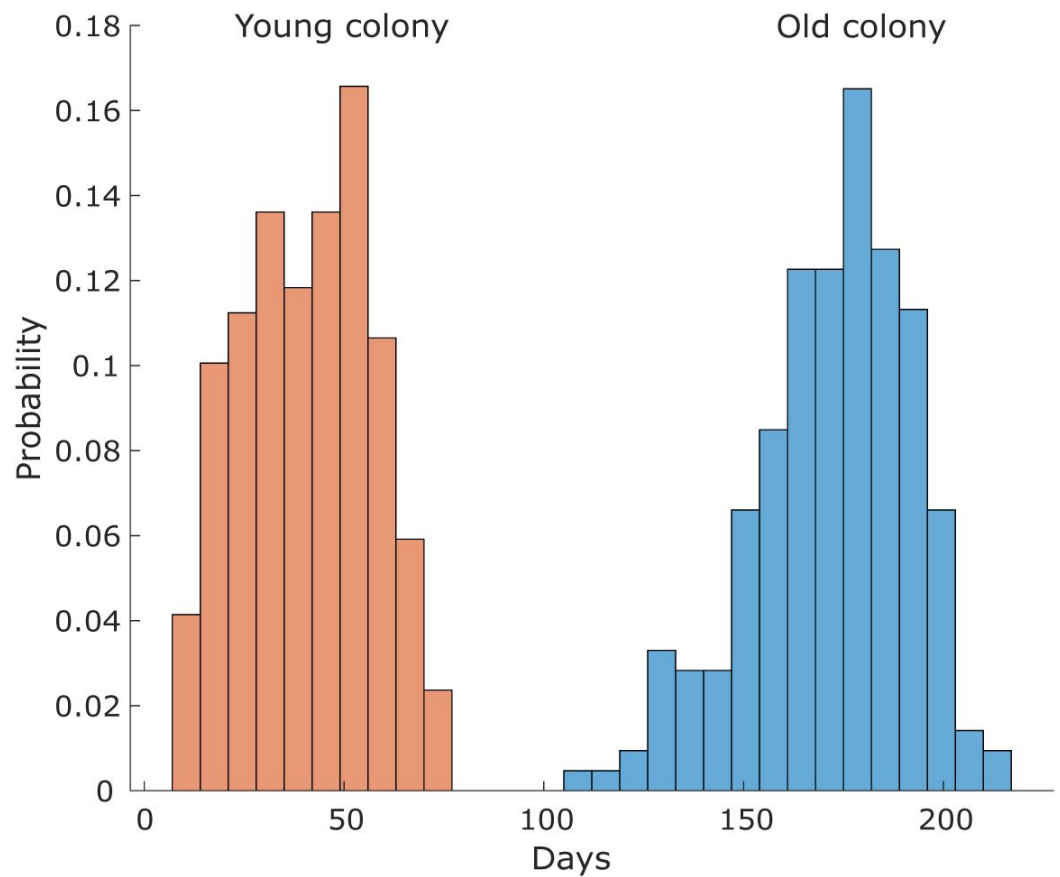
SI Figure 3.

Step-like relationship between excavated area and population growth.

A. The excavated area is plotted against their corresponding population sizes across specific colony maturation experiments where the step pattern was most evident. We can see a “step” like the relationship between the variables.

(B, C, D, E). Shows the change in area and population across experimental duration across the four specific nests. Peaks correspond to the increase in area or population.

Age distribution in fixed demographics experiments

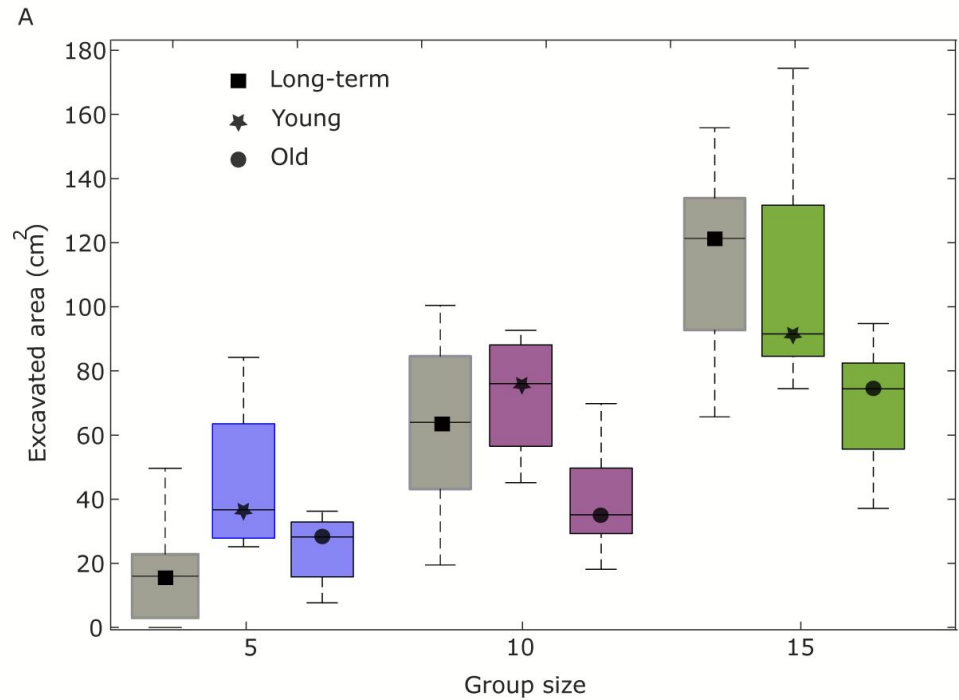


SI Figure 4.

Age distribution from the young and old colonies.

The age distribution of workers used for the fixed demographics young and old experiments is shown as a histogram. The ants in the fixed demographics young experiments had an average age of 40 ± 16 days, and old experiments had an average age of 171.56 ± 20 days. Errors represent the standard deviation.

Fixed demographics young colonies are similar to colony maturation experiments

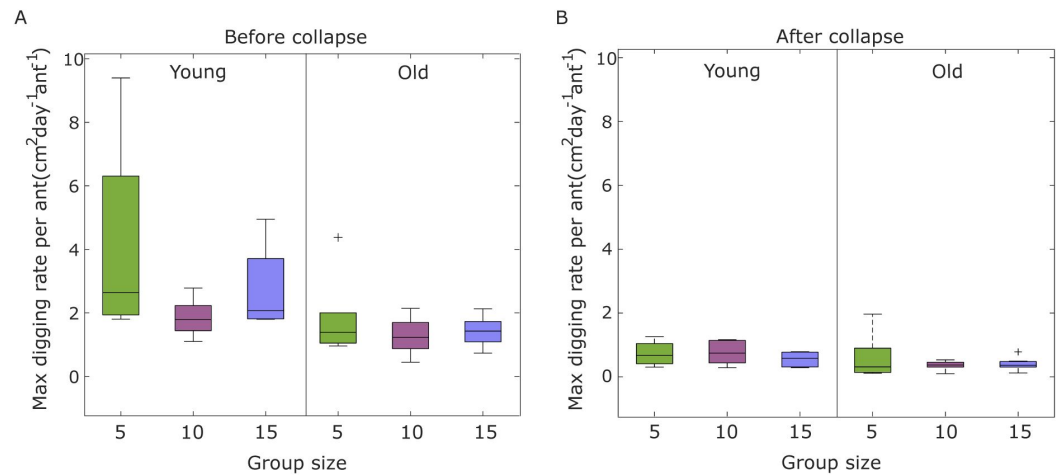


SI Figure 5.

Maximum excavated area from the fixed demographics excluding the area excavated by the queen.

A. The area excavated by the queen from the colony maturation experiments was excluded, and compared to the resultant area from the fixed demographics young, and old experiments. We find that the area excavated from the colony maturation experiments is similar to the fixed demographics young experiments.

Maximum digging rate from fixed demographics experiments

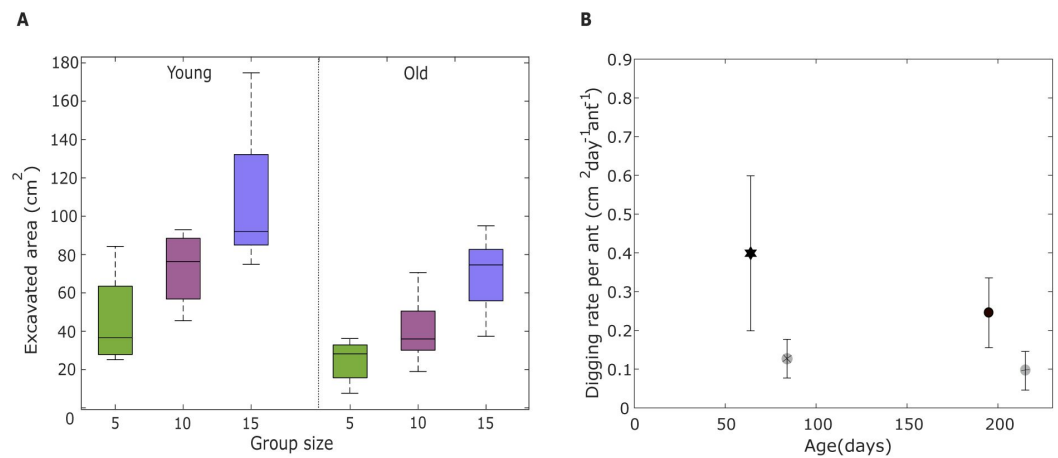


SI Figure 6.

Maximal digging rate before and after collapse.

The maximum digging per ant before (A) and after (B) the manual collapse events from the fixed demographics experiments. The maximum digging rate per ant is calculated by normalizing the maximum digging rate (change in area per day) by the number of ants. We find that there is no significant difference before and after collapse events (Kruskal-Wallis test, $\chi^2(5, 68) = 6.01, p = 0.30$)

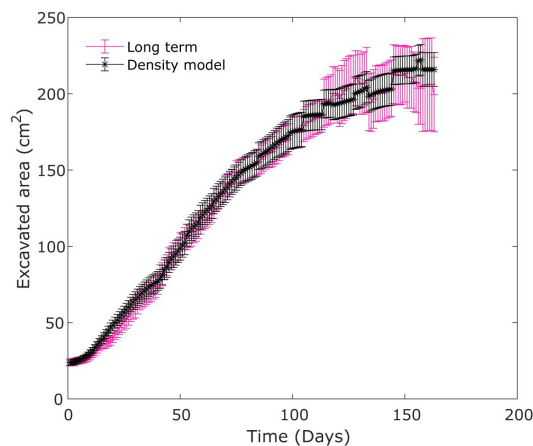
Excavated area, and digging rate from fixed demographics experiments



SI Figure 7.

Excavated area and digging rate.

A. The excavated area across group sizes from the fixed demographics young, and old experiments. We performed a linear regression on the excavated area across different group sizes for the fixed demographics young ($y = 6.35x + 0.73$, $R^2 = 0.43$), and old ($y = 4.57x - 0.73$, $R^2 = 0.57$). The positive slope indicates that the excavated area increases with group size for both the young, and old colonies. (B). The digging rate per ant for different age categories is shown. We find that the digging rate per ant is age-independent. Error bars represent the SEM.



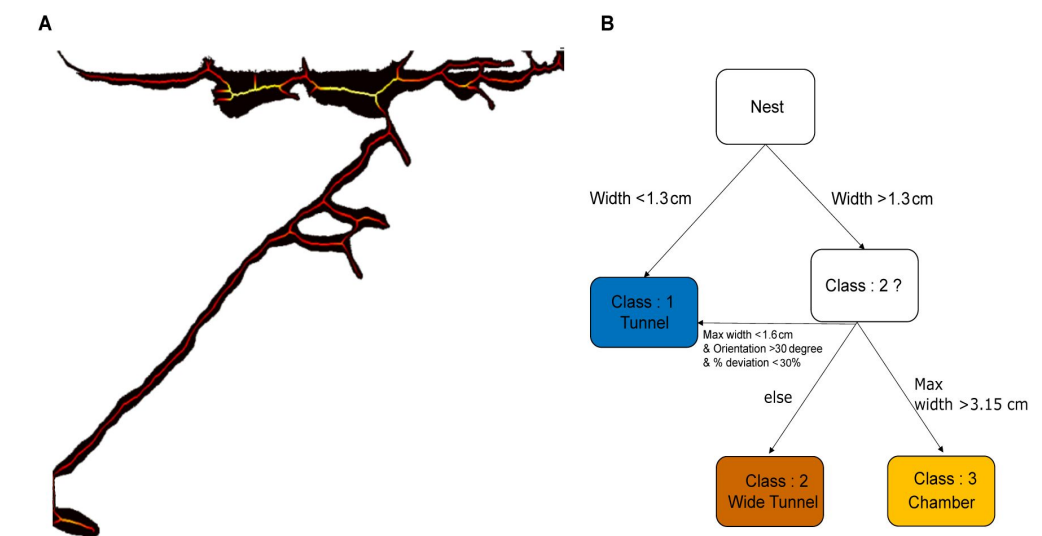
SI Figure 8.

Density model.

The progression of the excavated area based on the constant density regulation model, and its comparison to the colony maturation experiments. The constant density was fixed to be at $11.6 \text{ cm}^2 \text{ ant}^{-1}$. Error bars represent the SEM.

Nest skeletonization, segmentation and orientation

To estimate the total length of the nests, we created skeleton images of the binary images (SI. Fig 9 A). Initially, images were preprocessed with an averaging filter to smooth the shape edges. The skeleton was derived by MATLAB's, 'bwmorph' function using the 'thinning' morphological operator set to infinity. We further performed a 'Medial Axis Transform' (MAT) on the nest skeleton, where every skeleton pixel is assigned a distance value to the closest boundary of the excavated area. Many natural nests can be clearly divided into chambers and tunnels, while in other species the division is unclear (52). Moreover, previous studies defines chambers as flat horizontal structures compared to vertical narrow tunnels. With no available data on the architecture of natural nests of *Camponotus fellah* ants to compare with, we segmented the nest into tunnels, wide tunnels, and chambers using the decision algorithm as described in (SI. Fig 9 B). On the inverse of the binary image, we assigned each pixel in the excavated area to its shortest distance to the edge. Those assigned with a distance of more than 10 pixels ($\approx 0.65\text{cm}$) were designated as seeds for large chambers. Next, a geodesic distance transform assigned pixels the distances to these seeds and outlined the chambers. This created separate blobs, groups of connected pixels in a binary image, for the two classes with a width threshold of 1.30 cm between tunnels and potential chambers. From the width distribution, we choose 3.15 cm and 1.6 cm as high and low thresholds. Blobs above the high threshold were assigned as chambers. Blobs under the low threshold and relatively flat (absolute value of orientation under 30°) and overall regular in shape are added to the first class of tunnels. The residual blobs were defined as wide tunnels. The shape regularity was estimated by blobs with less than 30 % deviation from the mean of the MAT pixels. The resultant segmented images from the colony maturation, fixed demographics young, and old colonies are provided in (Fig. 5 A, B, and C). The orientation analysis was performed to quantify the angular orientation of all tunnels, wide tunnels, and chambers from the initial seeding point concerning the direction of gravity.

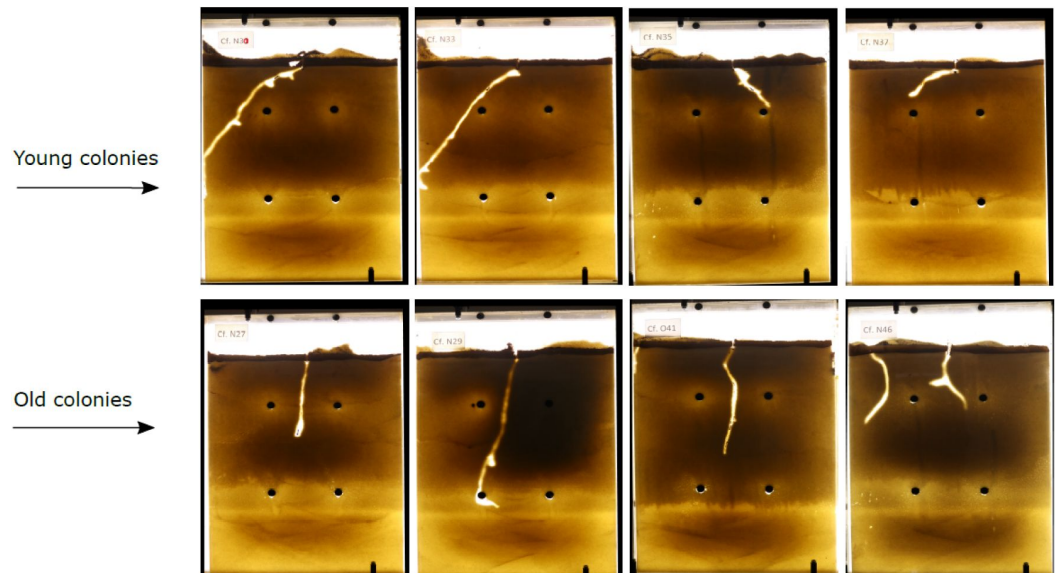


SI Figure 9.

Nest segmentation algorithm.

(A) Nest skeletonization - The medial-axis transform (MAT) image of the excavated area of a sample nest from the colony maturation experiments. The warmer color of the skeleton represents a larger distance to the edge of the excavated structure, shown in black. (B) The decision tree used for the structural segmentation of the nest images into three classes: namely, (1) Tunnels, (2) Wide tunnels, and (3) Chambers.

Nest architecture



SI Figure 10.

Nests excavated by colonies from the fixed demographics treatments.

The nest structures excavated by the young and old colonies from the fixed demographics experiments are shown. The first row corresponds to the nest images from the young colonies, and the second corresponds to the old.

Colony maturation, and fixed demographics experiments

Population change in the colony maturation experiments

We quantified the population growth in colony maturation experiments. Like the nest area, the population growth among the nest varied highly. In some experiments, only one or two workers eclosed ($n = 5$), while in others, the colony grew more than 15 workers ($n = 12$). The colony growth increased gradually in all nests until it saturated at a certain level. On average, the population grew logistically, reaching a plateau at roughly 12 ants after around 100 days. We fitted a classical population growth model to describe the population kinetics:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right) \quad (\text{Eq. 4})$$

On solving the equation, we get,

$$N(t) = \frac{K}{1 + (K-1)e^{-rt}} \quad (\text{Eq. 5})$$

We used $K = 19.0$, the maximal number of ants, and $r = -0.032$, to describe the population growth. We find that the logistic model can describe the population growth relatively well ($R^2 = 0.98$).

Age quantification from colony maturation experiments

During the nest photography session, we meticulously recorded the population of ants inhabiting each nest. By analyzing these population records, we were able to construct a comprehensive age demographics profile for the colony. For instance, consider the population change in the following nest:

On day 10, the population remained constant at 1 ant, with the ant being 10 days old. However, on day 11, the population increased by one, indicating the birth of a new ant. Therefore, on day 11, the newly born ant was 1 day old, while the existing ants were 11 days old. Subsequently, we analyzed the rest of the time points, allowing us to estimate the age distribution of ants in the entire nest.

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Reviewer #1 (Public review):

This study investigates how ant group demographics influence nest structures and group behaviors of *Camponotus fellah* ants, a ground-dwelling carpenter ant species (found locally in Israel) that build subterranean nest structures. Using a quasi-2D cell filled with artificial sand, the authors perform two complementary sets of experiments to try to link group behavior and nest structure: first, the authors place a mated queen and several pupae into their cell and observe the structures that emerge both before and after the pupae eclose (i.e., "colony maturation" experiments); second, the authors create small groups (of 5, 10, or 15 ants, each including a queen) within a narrow age range (i.e., "fixed demographic" experiments) to explore the dependence of age on construction. Some of the fixed demographic instantiations included a manually induced catastrophic collapse event; the authors then compared emergency repair behavior to natural nest creation. Finally, the authors introduce a modified logistic growth model to describe the time-dependent nest area. The modification introduces parameters that allow for age-dependent behavior, and the authors use their fixed demographic experiments to set these parameters, and then apply the

model to interpret the behavior of the colony maturation experiments. The main results of this paper are that for natural nest construction, nest areas, and morphologies depend on the age demographics of ants in the experiments: younger ants create larger nests and angled tunnels, while older ants tend to dig less and build predominantly vertical tunnels; in contrast, emergency response seems to elicit digging in ants of all ages to repair the nest.

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Reviewer #2 (Public review):

I enjoyed this paper and the approach to examining an accepted wisdom of ants determining overall density by employing age polyethism that would reduce the computational complexity required to match nest size with population (although I have some questions about the requirement that growth is infinite in such a solution). Moreover, the realization that models of collective behaviour may be inappropriate in many systems in which agents (or individuals) differ in the behavioural rules they employ, according to age, location, or information state. This is especially important in a system like social insects, typically held as a classic example of individual-as-subservient to whole, and therefore most likely to employ universal rules of behaviour. The current paper demonstrates a potentially continuous age-related change in target behaviour (excavation), and suggests an elegant and minimal solution to the requirement for building according to need in ants, avoiding the invocation of potentially complex cognitive mechanisms, or information states that all individuals must have access to in order to have an adaptive excavation output.

The only real reservation I have is in the question of how this relationship could hold in properly mature colonies in which there is (presumably) a balance between the birth and death of older workers. Would the prediction be that the young ants still dig, or would there be a cessation of digging by young ants because the area is already sufficient? Another way of asking this is to ask whether the innate amount of digging that young ants do is in any way affected by the overall spatial size of the colony. If it is, then we are back to a problem of perfect information - how do the young ants know how big the overall colony is? Perhaps using density as a proxy? Alternatively, if the young ants do not modify their digging, wouldn't the colony become continuously larger? As a non-expert in social insects, I may be misunderstanding and it may be already addressed in the citations used.

In any case, this is an excellent paper. The modelling approach is excellent and compelling, also allowing extrapolation to other group sizes and even other species. This to me is the main strength of the paper, as the answer to the question of whether it is younger or older ants that primarily excavate nests could have been answered by an individual tracking approach (albeit there are practical limitations to this, especially in the observation nest setup, as the authors point out). The analysis of the tunnel structure is also an important piece of the puzzle, and I really like the overall study.

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Reviewer #3 (Public review):

Summary:

In this study, Harikrishnan Rajendran, Roi Weinberger, Ehud Fonio, and Ofer Feinerman measured the digging behaviours of queens and workers for the first 6 months of colony development, as well as groups of young or old ants. They also provide a quantitative model describing the digging behaviours and allowing predictions. They found that young ants dig more slanted tunnels, while older ants dig more vertically (straight down). This finding is

important, as it describes a new form of age polyethism (a division of labour based on age). Age polyethism is described as a "yes or no" mechanism, where individuals perform or not a task according to their age (usually young individuals perform in-nest tasks, and older ones foraging). Here, the way of performing the task is modified, not only the propensity to carry it or not. This data therefore adds in an interesting way to the field of collective behaviours and division of labour.

The conclusions of the paper are well supported by the data. Measurements of the same individuals over time would have strengthened the claims.

Strengths:

I find that the measure of behaviour through development is of great value, as those studies are usually done at a specific time point with mature colonies. The description of a behaviour that is modified with age is a notable finding in the world of social insects. The sample sizes are adequate and all the information clearly provided either in the methods or supplementary.

Weaknesses:

I think the paper is failing to take into consideration or at least discuss the role of inter-individual variabilities. Tasks have been known to be undertaken by only a few hyper-active individuals for example. Comments on the choice to use averages and the potential roles of variations between individuals are in my opinion lacking. Throughout the paper wording should be modified to refer to the group and not the individuals, as it was the collective digging that was measured. Another issue I had was the use of "mature colony" for colonies with very few individuals and only 6 months of age. Comments on the low number of workers used compared to natural mature colonies would be welcome.

<https://doi.org/10.7554/eLife.100706.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

*This study investigates how ant group demographics influence nest structures and group behaviors of *Camponotus fellah* ants, a ground-dwelling carpenter ant species (found locally in Israel) that build subterranean nest structures. Using a quasi-2D cell filled with artificial sand, the authors perform two complementary sets of experiments to try to link group behavior and nest structure: first, the authors place a mated queen and several pupae into their cell and observe the structures that emerge both before and after the pupae eclose (i.e., "colony maturation" experiments); second, the authors create small groups (of 5, 10, or 15 ants, each including a queen) within a narrow age range (i.e., "fixed demographic" experiments) to explore the dependence of age on construction. Some of the fixed demographic instantiations included a manually induced catastrophic collapse event; the authors then compared emergency repair behavior to natural nest creation. Finally, the authors introduce a modified logistic growth model to describe the time-dependent nest area. The modification introduces parameters that allow for age-dependent behavior, and the authors use their fixed demographic experiments to set these parameters, and then apply the model to interpret the behavior of the colony maturation experiments. The main results of this paper are that for natural nest construction, nest areas, and morphologies depend on the age demographics of ants in the experiments: younger ants create larger nests and angled tunnels, while older ants*

tend to dig less and build predominantly vertical tunnels; in contrast, emergency response seems to elicit digging in ants of all ages to repair the nest.

We sincerely thank Reviewer #1 for the time and effort dedicated to our manuscript's detailed review and assessment. The revision suggestions were constructive, and we will incorporate them into the next version to improve the manuscript.

Reviewer #2 (Public review):

I enjoyed this paper and the approach to examining an accepted wisdom of ants determining overall density by employing age polyethism that would reduce the computational complexity required to match nest size with population (although I have some questions about the requirement that growth is infinite in such a solution). Moreover, the realization that models of collective behaviour may be inappropriate in many systems in which agents (or individuals) differ in the behavioural rules they employ, according to age, location, or information state. This is especially important in a system like social insects, typically held as a classic example of individual-as-subservient to whole, and therefore most likely to employ universal rules of behaviour. The current paper demonstrates a potentially continuous age-related change in target behaviour (excavation), and suggests an elegant and minimal solution to the requirement for building according to need in ants, avoiding the invocation of potentially complex cognitive mechanisms, or information states that all individuals must have access to in order to have an adaptive excavation output.

We sincerely thank reviewer #2 for the time and effort dedicated to our manuscript's detailed review and assessment. The insightful feedback provided by the reviewer will be incorporated into the successive revisions.

The only real reservation I have is in the question of how this relationship could hold in properly mature colonies in which there is (presumably) a balance between the birth and death of older workers. Would the prediction be that the young ants still dig, or would there be a cessation of digging by young ants because the area is already sufficient? Another way of asking this is to ask whether the innate amount of digging that young ants do is in any way affected by the overall spatial size of the colony. If it is, then we are back to a problem of perfect information - how do the young ants know how big the overall colony is? Perhaps using density as a proxy? Alternatively, if the young ants do not modify their digging, wouldn't the colony become continuously larger? As a non-expert in social insects, I may be misunderstanding and it may be already addressed in the citations used.

We thank the reviewer for this interesting question. We find that the nest excavation is predominantly performed by the younger ants in the nest and the nest area increase is followed by an increase in the population. However, if the young ants dig unrestricted, this could result in unnecessary nest growth as suggested by reviewer #2. Therefore, we believe that the innate digging behavior of ants could potentially be regulated by various cues such as;

(a) Density-based: If the colony becomes less dense as its area expands, this could serve as a feedback signal for young ants to reduce or stop digging, as described in references (25, 29, 30).

(b) Pheromone depositions: If the colony reaches a certain population density, pheromone signals could inhibit further digging by young ants, references (25, 29,) or space usage as a proxy for the nest area.

Thus, rather than perfect information, decentralized control, and digging-based local cues probably regulate the level of age-dependent digging, without the ants needing to estimate the overall colony size or nest area.

In any case, this is an excellent paper. The modelling approach is excellent and compelling, also allowing extrapolation to other group sizes and even other species. This to me is the main strength of the paper, as the answer to the question of whether it is younger or older ants that primarily excavate nests could have been answered by an individual tracking approach (albeit there are practical limitations to this, especially in the observation nest setup, as the authors point out). The analysis of the tunnel structure is also an important piece of the puzzle, and I really like the overall study.

We thank the reviewer for the comments. We completely agree that individual tracking of ants within our experimental setup would have been the ideal approach, but we were limited by technical and practical limitations of the setup as pointed out by the reviewer such as;

(a) Continuous tracking of ants in our nests would have required a camera to be positioned at all times in front of the nest, which necessitates a light background. Since *Camponotus fellah* ants are subterranean, we aimed to allow them to perform nest excavation in conditions as close to their natural dark environment as possible. Additionally, implementing such a system in front of each nest would have reduced the sample sizes for our treatments.

(b) The experimental duration of our colony maturation and fixed demographics experiments extended for up to six months (unprecedented durations in these kinds of measurements). These naturally limited our ability to conduct individual tracking while maintaining the identity of each ant based on the current design.

Reviewer #3 (Public review):

Summary:

In this study, Harikrishnan Rajendran, Roi Weinberger, Ehud Fonio, and Ofer Feinerman measured the digging behaviours of queens and workers for the first 6 months of colony development, as well as groups of young or old ants. They also provide a quantitative model describing the digging behaviours and allowing predictions. They found that young ants dig more slanted tunnels, while older ants dig more vertically (straight down). This finding is important, as it describes a new form of age polyethism (a division of labour based on age). Age polyethism is described as a "yes or no" mechanism, where individuals perform or not a task according to their age (usually young individuals perform in-nest tasks, and older ones foraging). Here, the way of performing the task is modified, not only the propensity to carry it or not. This data therefore adds in an interesting way to the field of collective behaviours and division of labour.

The conclusions of the paper are well supported by the data. Measurements of the same individuals over time would have strengthened the claims.

We sincerely thank reviewer #3 for the time and effort dedicated to our manuscript's detailed review and assessment. We completely agree with the reviewer's comments on the measurements of the same individuals over time, however, we were limited by the technical and experimental limitations as described above and pointed out by reviewer #2.

Strengths:

I find that the measure of behaviour through development is of great value, as those studies are usually done at a specific time point with mature colonies. The description of

a behaviour that is modified with age is a notable finding in the world of social insects. The sample sizes are adequate and all the information clearly provided either in the methods or supplementary.

We thank the reviewer #3 for this assessment.

Weaknesses:

I think the paper is failing to take into consideration or at least discuss the role of inter-individual variabilities. Tasks have been known to be undertaken by only a few hyper-active individuals for example. Comments on the choice to use averages and the potential roles of variations between individuals are in my opinion lacking. Throughout the paper wording should be modified to refer to the group and not the individuals, as it was the collective digging that was measured. Another issue I had was the use of "mature colony" for colonies with very few individuals and only 6 months of age. Comments on the low number of workers used compared to natural mature colonies would be welcome.

Regarding main comment 1

We completely agree with the reviewer's comment on considering inter-individual variability based on activity levels. We have discussed how individual morphological variability could influence digging behavior (references: 28, 31), and we will elaborate further on this aspect in future revisions.

Regarding main comment 2:

We agree with the reviewer's comments regarding the wording. The term "mature colony" will be revised in future versions. The wording ("mature colony") will be changed and addressed in the future revisions. We were practically limited by the continuation of the experiments for more than 6 months of age predominantly due to the stability of nests as they were made with a sand-soil mix. We also acknowledge that the colony sizes attained in our maturation experiments may be smaller than those of naturally matured colonies. This trend was observed generally in lab-reared colonies and could be attributed to differences in microclimatic conditions, foraging opportunities, space availability, and other factors. We will address these aspects in more detail in future revisions.

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